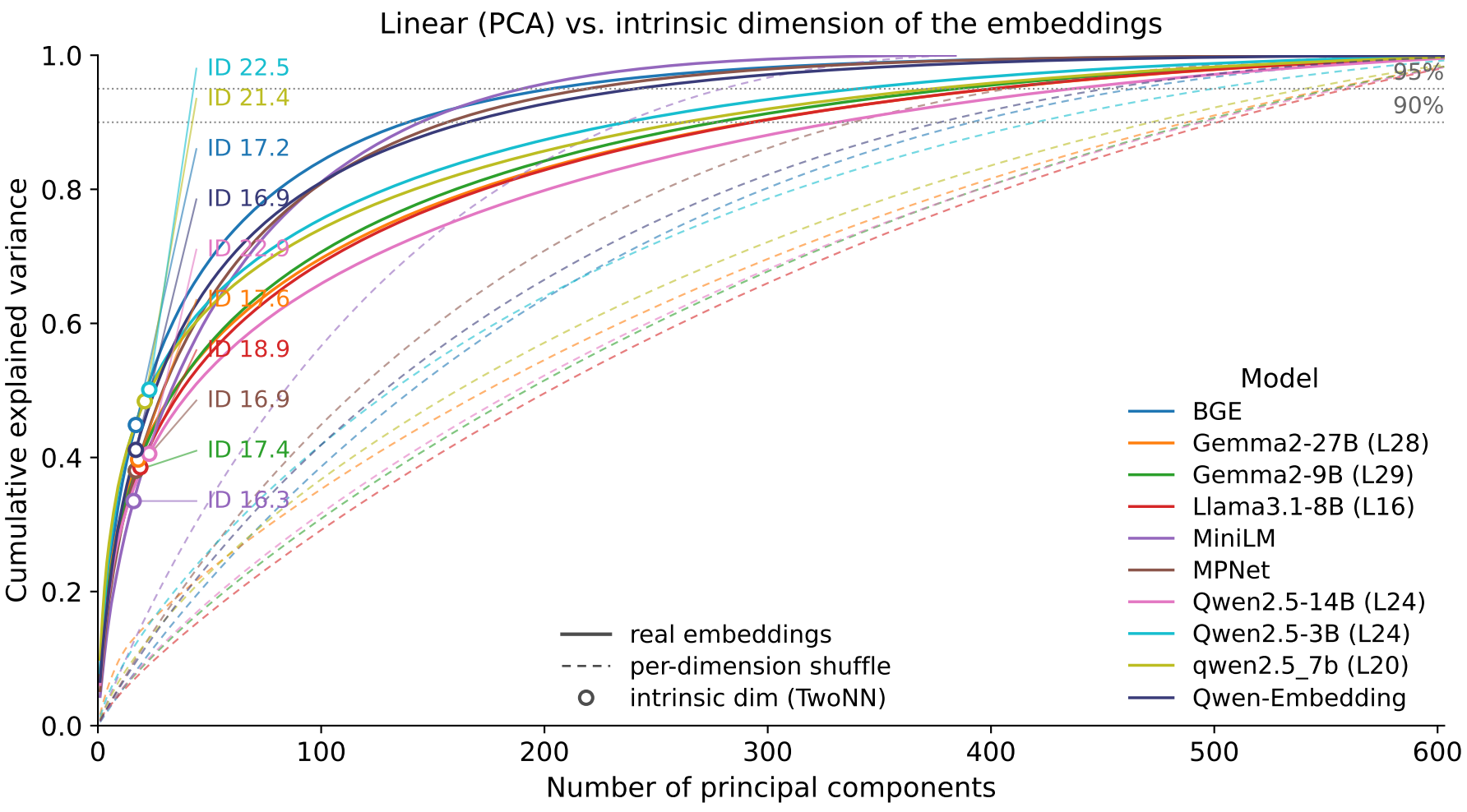
Linear (PCA) vs. intrinsic dimension of the embeddings
Cumulative explained variance
Number of principal components
1.0
0.8
0.6
0.4
0.2
0.0
0
100
200
300
400
500
600
95%
90%
ID 22.5
ID 21.4
ID 17.2
ID 16.9
ID 22.9
ID 17.6
ID 18.9
ID 16,9
ID 17.4
ID 16.3
real embeddings
per-dimension shuffle
intrinsic dim (TwoNN)
Model
BGE
Gemma2-27B (L28)
Gemma2-9B (L29)
Llama3.1-8B (L16)
MiniLM
MPNet
Qwen2.5-14B (L24)
Qwen2.5-3B (L24)
qwen2.5_7b (L20)
Qwen-Embedding